

Creating your first Java program

Let us get our hands dirty with Eclipse. In this section, we'll write the traditional 'Hello World' program in Eclipse. To create a new workspace, just launch Eclipse. It will ask for the name and location of the workspace. The workspace location on your machine is where the application's code will be stored.

Let us create our first Java project. Navigate to *File->New->Other*, then select *Java Project* from the drop down list and follow the on-screen directions to create the project.

I've created the HelloWorld project, which we can see in the 'Package Explorer'.

Let us create a Java package. A package is a logical container that contains related Java classes. To create a package, select the 'src' directory from the 'Package Explorer'. Navigate to *File->New->Package*, provide the package's name and click on the 'Finish' button. I've used 'com.osfy' as a package name.

Now let us create a Java class which will contain actual code. To create a class, select the newly created package from the 'Package Explorer'. Navigate to *File->New->Class*, provide the class name and click on the 'Finish' button. This action will open a class file in the editor. After adding the *main()* method, the code will look like what's shown in Figure 4.

Now let us compile and run this Java class. We can do so as a Java application. Select *Class* from the *Package Explorer->right click ->Run As->Java Application*. If everything goes well, it will show an output in the console window as seen in Figure 3.

If there are any errors, these will be listed under the 'Problems' tab.

Debugging a Java class

The debugger is a programmer's best friend. Using it, we can stop the execution of a program at a certain point to examine variables, step into methods and do many other tasks. In this section, we'll look at how to debug

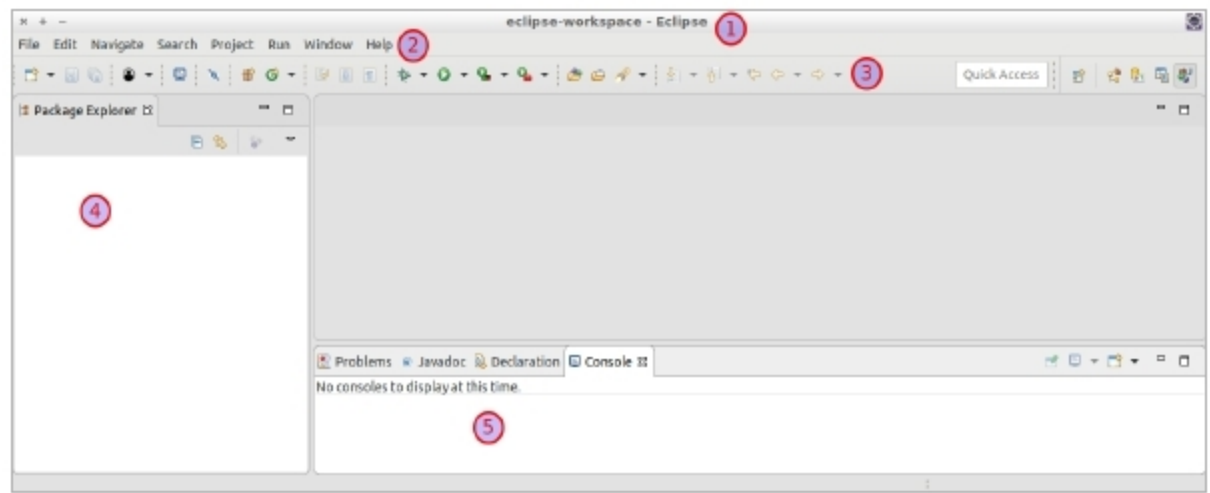


Figure 1: Eclipse window

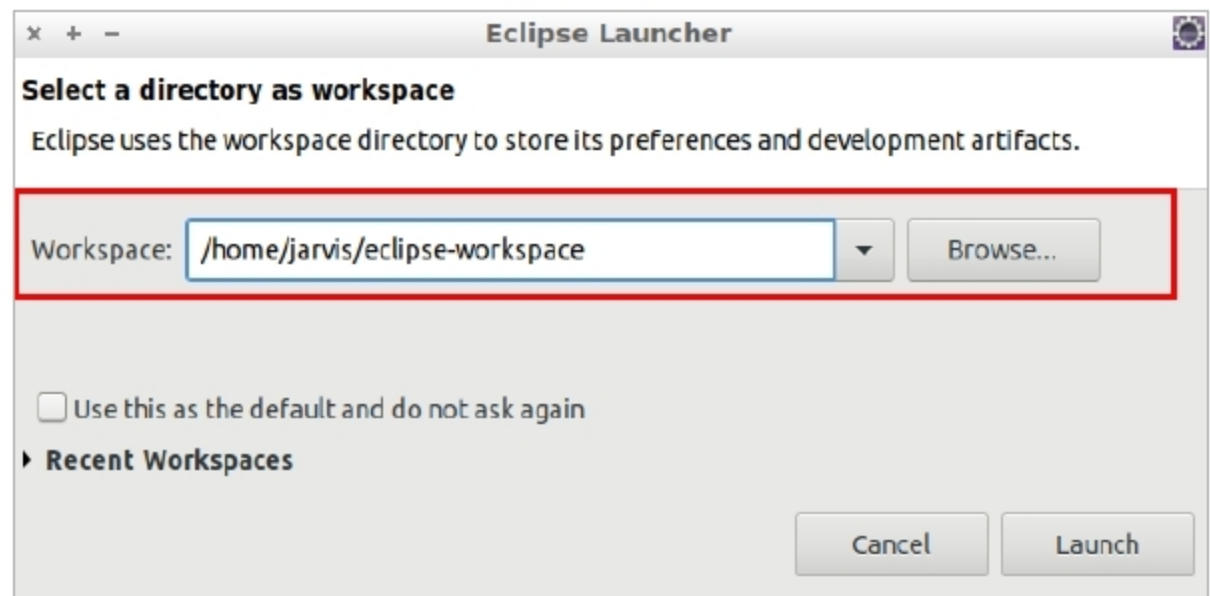


Figure 2: The Eclipse workspace

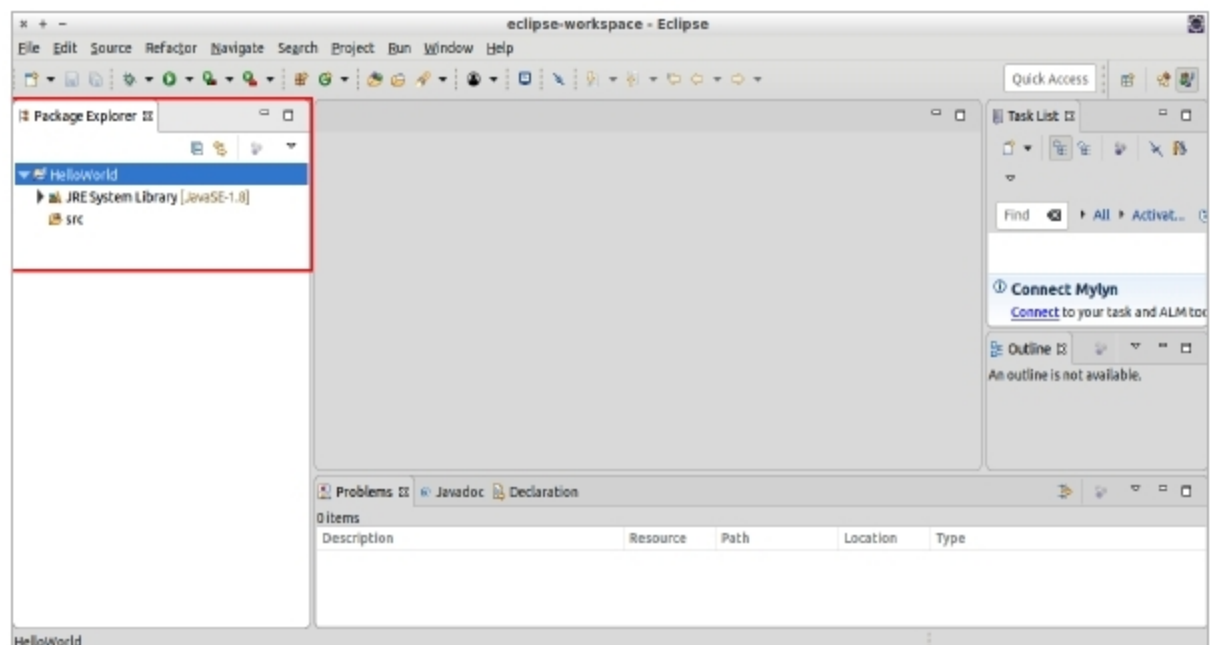


Figure 3: Eclipse project

a simple Java class.

Let us create a Java class with the following code:

```
package com.osfy;
public class AdditionExample {
    public static void main(String[]
args) {
    int a, b, c;
```

```
a = 10;
b = 20;
c = a + b;
```

```
System.out.println(a + " + " +
b + " = " + c);
}
```

We can stop the execution of a program by setting breakpoints. To set